JLPT AI App, Progress Report No. 10, HP

Padauk Lab

Report	Period	Date	Prepared by
No. 10	Oct 26 to Oct 30, 2026	Oct 30, 2026	HP (Backend and Infrastructure)

1. Summary

The app runs on a real iPhone, with a real online server. The backend is online for $0 (Render + Neon + UptimeRobot), and the whole team can use it from any network. The app now has **10 sessions** (100 units, 2,000 core N1 words). Sessions 1 to 3 are complete (600 words), and Session 1 has both quizzes. The pass prediction and the AI study tip are built. A security test on the online server found a real weakness, and it is fixed. **The biggest blocker now: the Session 1 mock test.** Without it, nobody can open Session 2.

2. Completed by HP

No.	Task	Status
1	Change password (#57), "More examples" with a daily limit (#65), forgot password with a 6-digit code (#66)	Merged
2	Library updates (#70), wave dash fix (#72), shared question rules (#73)	Merged
3	Daily study reminder as a push notification (#74)	Merged
4	10 sessions instead of 20 (#76)	Merged
5	Backend online: Render (server), Neon (database), UptimeRobot (keeps it awake), all free (#71, #80)	Live
6	Security fix: the login limit now sees each user's real address behind Cloudflare (#79)	Merged and tested online
7	Pass prediction from mock tests, contract v0.8 (#95)	Merged
8	AI study tip connected to the study plan (#97)	In review
9	Loaded 30 units and the Session 1 quizzes into the online database	Done
10	Gemini key on the server (only on Render, never in code), web testing allowed (CORS)	Done
11	Expo organization `padauk-lab`, so the app belongs to the team, not one person	Done
12	Reviewed team work: #69, #78, #81 to #94, #96	Done

Automated tests: 513 on main today, 525 when #97 is merged.

3. Security at a glance (new this time)

Protection	What it stops
The login limit reads the address Cloudflare writes itself (`cf-connecting-ip`), which nobody can fake. Tested online: 31 tries with fake addresses end with 429	Guessing passwords on many accounts by faking addresses
Secrets only on Render's Environment page (encrypted). A separate Gemini key just for the server	Keys in GitHub, chat or the app
The AI key is never in the app: the app asks our server, the server asks Gemini	Anyone reading the key out of the app
Web access only from the team's own computers (`CORS_ORIGINS`)	Other websites using our server
The pass prediction and the study tip are calculated on the server	Faking a better result in the app

4. Team highlights

Member	What they did	Status
LMHP	Saved words notebook (#69), Expo SDK 58 (#78), ox cart loading screen (#83)	Merged
LMHP	Quiz and "Why?" connected to the real backend (#94)	Merged
SWYH	`plan_tip()`, the AI study tip, with checks that the tip is Burmese and has no numbers (#96)	Merged
ANK	Session 1 Quiz 1 and Quiz 2, 80 questions (#82). Agreed the pass prediction formula	Merged
Deedeedar	Session 2 Units 4 to 10 (#81), Session 3 Units 1 to 10 (#84 to #93)	Merged

5. Content status

Session	1	2	3	4 to 10
Units	10 of 10	10 of 10	10 of 10	Not yet
Quizzes	Done	Not yet	Not yet	Not yet
Mock test	**Missing**	Not yet	Not yet	Not yet

- 30 of 100 units (30%), 600 words, 30 grammar patterns. No repeated words or grammar.
- **The Session 1 mock test is the most urgent task.** Every user stops at the end of Session 1 without it.
- No proverb files yet. The format and loader are ready.

6. Decisions made

Area	Decision
App size	10 sessions, 100 units, 2,000 core N1 words for learners who already have N2
Hosting	$0: Render, Neon, UptimeRobot. The free server is kept awake
Expo	Stay on SDK 58 (don't go back). The app belongs to the Expo organization `padauk-lab`
Pass prediction	Latest 2 mock tests, first tries only, newer one 60%. Score = 70% language knowledge + 30% reading. 60% = 50 chance, 2.5 per point, between 5 and 95. A section under 32% caps it at 15, with a warning. No listening
Mock test reading	1, 3 and 4 questions (short, medium, long)
Study tip	Written after the plan is saved, wait at most about 8 seconds, never with numbers
Email check at sign-up	Not now. Add email verification before the public release

7. Action needed (everyone)

After pulling main:

```
cd frontend && npm install
docker compose run --rm --build backend alembic upgrade head
```

To test against the online server, in `frontend/.env.local`:

```
EXPO_PUBLIC_API_URL=https://kyugo-api.onrender.com/api/v1
```

Then start Expo with the cache cleared (`npx expo start -c`) and **sign up a new account**: the online database is separate from your laptop.

8. Next tasks

No.	Task	Owner
1	**Session 1 mock test** (`docs/mock-test-format.md`, reading 1, 3 and 4)	ANK, Deedeedar
2	Quiz 1 and Quiz 2 for Sessions 2 and 3	ANK, Deedeedar
3	Connect the mock test, "More examples", forgot and change password. Show the pass prediction warning and the no-listening note	LMHP
4	A 15-second time limit on app requests, so a bad connection shows a message instead of spinning	LMHP
5	Ask for notification permission and register the phone (#33)	LMHP
6	Review and merge #97, Report 11	HP

9. Open decisions

1. Proverb of the day notification: what time to send it
2. Our own email domain, so reset codes really reach inboxes (needed before the public release)
3. When to get the Apple Developer membership ($99 a year, needed to publish)
4. Shuffle the answer choices in quizzes (later, small)
5. Add a Japanese definition field (`meaning_ja`) to the contract? (team)
6. Keep `frontend/.claude/settings.json` (an AI tool setting) in the repo? (team)